

Laboratory Report

EIT & MEC – Digi Lab 1: Switching Circuits

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Abstract

This report documents the experimental investigation of monostable and astable multivibrators as well as a quartz crystal oscillator, conducted in Digi Lab 1 at Fachhochschule Vorarlberg on 21/22 April 2026. Four circuits were assembled and analysed: a monostable multivibrator based on the NE555 timer IC, an astable multivibrator also using the NE555, an astable oscillator built from two CMOS inverters (74C00), and a 12 MHz quartz crystal oscillator. For each circuit, theoretically calculated values for pulse duration and oscillation period were compared against oscilloscope measurements, and deviations were discussed with respect to component tolerances and measurement equipment effects. *[Refine after experiments are completed – max. 200 words.]*

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1 Introduction

Switching circuits are fundamental building blocks in digital and mixed-signal electronics. This laboratory exercise focuses on two important classes of such circuits: multivibrators, which generate or shape rectangular pulse signals, and crystal oscillators, which provide highly stable reference frequencies.

A **monostable multivibrator** (one-shot) produces a single output pulse of defined duration in response to a trigger event. An **astable multivibrator** oscillates continuously between two states without any external trigger, generating a periodic square wave. Both types can be realised with the NE555 timer IC, whose internal structure consists of two comparators, a flip-flop, a discharge transistor, and an output stage.

The **NE555** is a versatile and widely used timer IC. In monostable mode, the output pulse width is determined solely by an external RC network according to $t_H = 1.1 RC$. In astable mode, the frequency and duty cycle depend on two resistors and one capacitor.

As a complementary approach, an astable oscillator can also be constructed from two cascaded CMOS inverter gates with a feedback RC network. Finally, a quartz crystal oscillator exploits the piezoelectric resonance of a quartz crystal to achieve frequency stabilities several orders of magnitude better than RC-based circuits.

The objectives of this laboratory are:

- To understand and verify the operating principle of monostable and astable multivibrators using the NE555 IC.
- To compare calculated and measured values for pulse duration and oscillation period, and to quantify and explain deviations.
- To investigate the influence of the measurement setup (oscilloscope probes vs. BNC cables) on the circuit behaviour.
- To analyse the astable multivibrator realised with two CMOS inverters.
- To observe and explain the behaviour of a quartz crystal oscillator under varying supply voltages.

The theoretical background was reviewed according to [1] and [2].

2 Equipment and Materials

All circuits were assembled on a standard experimentation (breadboard) system. Signals were observed and recorded with a digital storage oscilloscope. Table 1 lists the equipment used.

Table 1: Equipment and components used in this experiment.

Item	Specification / Part No.	Asset / S/N
Digital oscilloscope	<i>[Model, e.g. Rigol DS1054Z]</i>	<i>[S/N]</i>
Oscilloscope probe (x10)	<i>[Model]</i>	<i>[S/N]</i>
BNC cable (50 Ω)	<i>[Model]</i>	—
DC power supply	10 V / 5 V	<i>[S/N]</i>
Experimentation board	—	—
NE555 timer IC	NE555	—
CMOS NAND IC	74C00	—
Quartz crystal	12 MHz	—
Resistors	see individual tasks	—
Capacitors	see individual tasks	—

3 Theoretical Background

3.1 NE555 Internal Structure

The NE555 contains three equal resistors forming a voltage divider that sets the reference levels at $\frac{1}{3}V_{CC}$ and $\frac{2}{3}V_{CC}$, two comparators, an SR flip-flop, an npn discharge transistor (pin 7), and a push-pull output stage (pin 3).

3.2 Monostable Mode

In monostable mode, the capacitor C charges through R_1 after a falling trigger edge at pin 2 pulls the output high. The output returns low once V_C reaches $\frac{2}{3}V_{CC}$, which gives the pulse duration:

$$t_H = 1.1 \cdot R_1 \cdot C \quad (1)$$

3.3 Astable Mode

In astable mode, C charges through $R_1 + R_2$ and discharges through R_2 alone. The resulting high and low times are:

$$t_H = 0.693 (R_1 + R_2) C \quad (2)$$

$$t_L = 0.693 R_2 C \quad (3)$$

$$T = 0.693 (R_1 + 2R_2) C \quad (4)$$

3.4 RC Oscillator with CMOS Inverters

Two cascaded inverters with a series resistor R and capacitor C in the feedback path form a relaxation oscillator. The threshold voltages of the CMOS gates determine the switching points, giving an oscillation frequency approximately:

$$f \approx \frac{1}{2.2 RC} \quad (5)$$

3.5 Quartz Crystal Oscillator

A quartz crystal behaves as a very high-Q series/parallel resonant circuit. When connected in the feedback loop of an inverter amplifier, it forces oscillation at its resonant frequency, providing frequency stability in the range of $\pm 10^{-5}$ to $\pm 10^{-6}$.

4 Procedures

All experiments were carried out according to the laboratory instruction sheet *Lab_01_SwitchingCircuits*, Rev. 1.1, dated 21/22.04.2026. Circuits were assembled on the experimentation board. A supply voltage of 10 V was used for tasks 3.1–3.3, and 5 V for task 3.4 (quartz oscillator).

Oscilloscope measurements were performed in two configurations where applicable:

1. Using 10:1 oscilloscope probes (input impedance $10\text{ M}\Omega \parallel 10\text{ pF}$).
2. Using direct $50\text{ }\Omega$ BNC cables (task 3.1.4 only).

Screenshots were saved for each relevant waveform. Component values were verified with a multimeter before assembly where possible.

5 Results and Discussion

5.1 Task 3.1 – Monostable Multivibrator (NE555)

Component values used: $C = 1\text{ }\mu\text{F}$, $R_1 = 1\text{ M}\Omega$, $V_{CC} = 10\text{ V}$.

5.1.1 Task 3.1.1 – Calculation of Pulse Duration

Applying equation (1):

$$\begin{aligned} t_{H,\text{calc}} &= 1.1 \times 1\text{ M}\Omega \times 1\text{ }\mu\text{F} \\ &= 1.1\text{ s} \end{aligned}$$

$t_{H,\text{calc}} = 1.1\text{ s}$

5.1.2 Task 3.1.2 – Measurement with Oscilloscope Probes

Both the output signal y (pin 3) and the capacitor voltage V_C were measured simultaneously using 10:1 oscilloscope probes.

Table 2: Measured values – monostable multivibrator, oscilloscope probes.

Quantity	Symbol	Calculated	Measured
Pulse duration	t_H	1.1 s	?? s
Peak capacitor voltage	$V_{C,\text{max}}$	6.67 V	?? V
Rise time (10 %–90 %)	t_r	—	?? ms



Figure 1: Output y (pin 3, upper trace) and capacitor voltage V_C (lower trace) measured with 10:1 probes – monostable multivibrator.

5.1.3 Task 3.1.3 – Explanation of Differences

Hints for this section

- Resistor tolerance: $\pm 5\%$; capacitor tolerance: $\pm 20\%$
- NE555 internal resistance on pin 7 is not zero ($\approx$ a few Ω – 100Ω)
- Comparator threshold not exactly $\frac{2}{3}V_{CC}$ (data sheet: $\pm 15\%$)
- Leakage current of capacitor shifts the effective time constant
- Probe capacitance (10 pF) negligible for $1\text{ }\mu\text{F}$

The measured pulse duration deviates from the calculated value by $\Delta t_H = ??\text{ s}$, corresponding to a relative error of $\delta = ??\%$.

[Discuss which tolerance dominates (likely capacitor) and whether the deviation is within the expected tolerance band.]

5.1.4 Task 3.1.4 – Measurement with BNC Cables

The measurement from task 3.1.2 was repeated, replacing the oscilloscope probes with direct BNC cables (50Ω input impedance).

Table 3: Comparison: oscilloscope probes vs. BNC cables – monostable multivibrator.

Quantity	Calculated	Probes	BNC cable
Pulse duration t_H	1.1 s	?? s	?? s
Peak $V_{C,\max}$	6.67 V	?? V	?? V

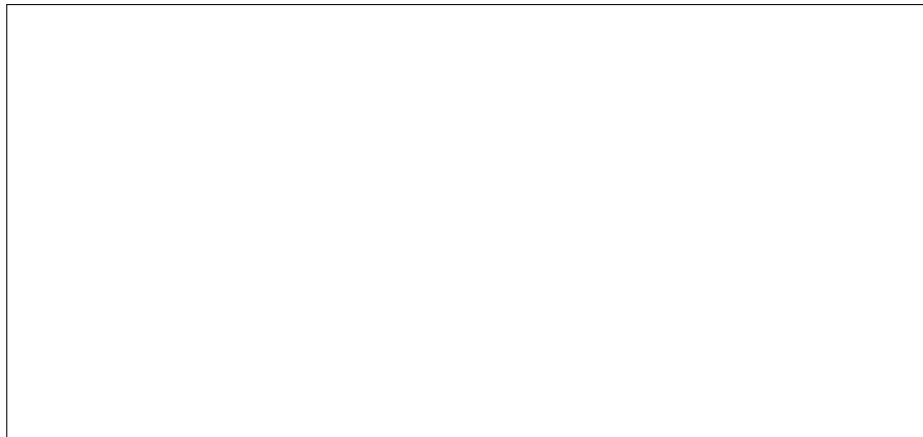


Figure 2: Output y and V_C measured via BNC cable – monostable multivibrator.

Hints for this section

- BNC cable: $50\,\Omega$ oscilloscope input, vs. $10\,\text{M}\Omega$ for probes
- The $50\,\Omega$ load in parallel with $R_1 = 1\,\text{M}\Omega$ is negligible at the output, but loading the capacitor node changes the effective discharge path
- Capacitor discharges faster through the low BNC impedance – shorter t_H
- Amplitude of V_C is attenuated by the voltage divider formed by R_1 and the $50\,\Omega$ load

[Describe and explain the observed differences. Calculate the effective parallel resistance and the expected new time constant if relevant.]

5.1.5 Task 3.1.5 – Output-Only Measurement (No Capacitor Probing)

The output signal y at pin 3 was measured alone, without a probe on the capacitor node.

Table 4: Output pulse duration with and without simultaneous capacitor probing.

Quantity	With V_C probed	Without V_C probed
Pulse duration t_H	?? s	?? s

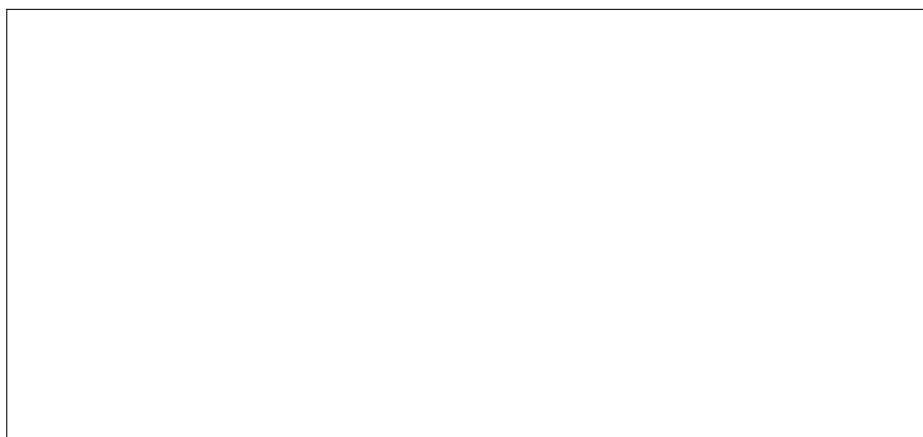


Figure 3: Output y (pin 3) only – monostable multivibrator, no capacitor probing.

Hints for this section

- Probe capacitance (10 pF) adds to C at the capacitor node
- Removing the probe from the capacitor node slightly decreases total capacitance, resulting in a slightly shorter t_H
- Effect is very small: $10 \text{ pF} \ll 1 \text{ }\mu\text{F}$
- Output stage (pin 3) is low impedance, so probing there has negligible effect

[Quantify any observed difference and state whether it is within measurement uncertainty.]

5.2 Task 3.2 – Astable Multivibrator (NE555)

Component values used: $R_1 = 10 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, $C = 100 \text{ nF}$, $V_{CC} = 10 \text{ V}$.

5.2.1 Task 3.2.1 – Calculation of Oscillation Period

Applying equations (2)–(4):

$$t_H = 0.693 \times (10 \text{ k}\Omega + 10 \text{ k}\Omega) \times 100 \text{ nF} = 1.386 \text{ ms}$$

$$t_L = 0.693 \times 10 \text{ k}\Omega \times 100 \text{ nF} = 0.693 \text{ ms}$$

$$T = t_H + t_L = 2.079 \text{ ms}$$

$$f = \frac{1}{T} = 481.0 \text{ Hz}$$

$$D = \frac{t_H}{T} = \frac{R_1 + R_2}{R_1 + 2R_2} = 66.7 \%$$

$$T_{\text{calc}} = 2.079 \text{ ms}, \quad f_{\text{calc}} = 481 \text{ Hz}, \quad D_{\text{calc}} = 66.7 \%$$

5.2.2 Task 3.2.2 – Measurement

Table 5: Measured values – astable multivibrator (NE555).

Quantity	Symbol	Calculated	Measured
High time	t_H	1.386 ms	?? ms
Low time	t_L	0.693 ms	?? ms
Period	T	2.079 ms	?? ms
Frequency	f	481 Hz	?? Hz
Duty cycle	D	66.7 %	?? %
Capacitor max.	$V_{C,\text{max}}$	6.67 V	?? V
Capacitor min.	$V_{C,\text{min}}$	3.33 V	?? V



Figure 4: Output y (pin 3, upper trace) and capacitor voltage V_C (lower trace) – astable multivibrator, NE555.

5.2.3 Task 3.2.3 – Explanation of Differences

Hints for this section

- Resistor tolerance $\pm 5\%$, capacitor tolerance $\pm 20\%$
- Internal NE555 discharge transistor $V_{CE,sat} > 0$ raises effective R_2
- Comparator thresholds deviate from ideal $\frac{1}{3}$ and $\frac{2}{3}$
- Probe adds parasitic capacitance to the C node
- Since $R_1 = R_2$, the theoretical duty cycle is $\frac{2}{3}$; a diode in parallel with R_2 could yield 50 %

[State the absolute and relative deviations for T and f . Identify the dominant error source.]

5.2.4 Task 3.2.4 – Rising Edge Zoom

Table 6: Rising edge parameters – astable NE555 output.

Quantity	Symbol	Measured
Rise time (10 %–90 %)	t_r	?? μs
Overshoot	—	?? %

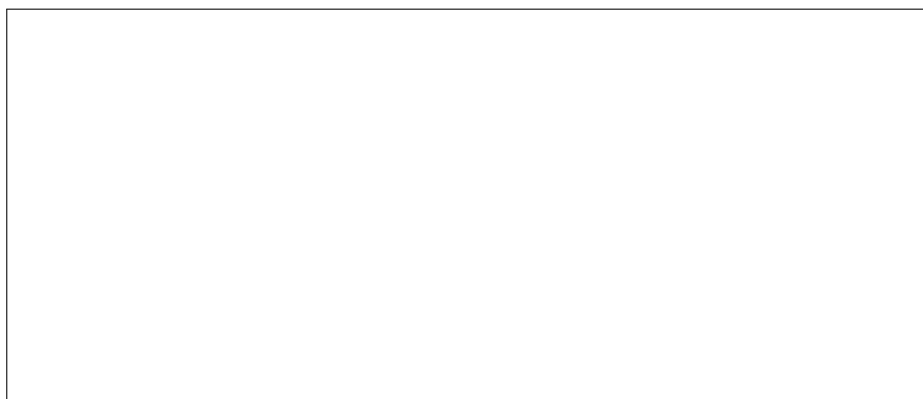


Figure 5: Zoomed rising edge of the NE555 astable output signal.

Hints for this section

- NE555 output stage is a BJT totem-pole; switching from saturation to cut-off takes finite time (storage time)
- Capacitive load at pin 3 (scope probe, wiring) slows the edge
- Possible overshoot/ringing due to parasitic inductance of wires
- Rising edge typically faster than falling edge due to asymmetric BJT structure

[Describe the waveform shape. Is there ringing or overshoot? Relate the observation to the internal BJT output stage and parasitic reactances.]

5.3 Task 3.3 – Astable Multivibrator Using Two Inverters (74C00)

Component values used: $R_S = 1\text{ k}\Omega$, $R = 10\text{ k}\Omega$, $C = 1\text{ nF}$, $V_{CC} = 10\text{ V}$.

Expected frequency from equation (5):

$$f_{\text{calc}} \approx \frac{1}{2.2 \times 10\text{ k}\Omega \times 1\text{ nF}} = 45.5\text{ kHz}$$

5.3.1 Task 1.1 – Circuit Function

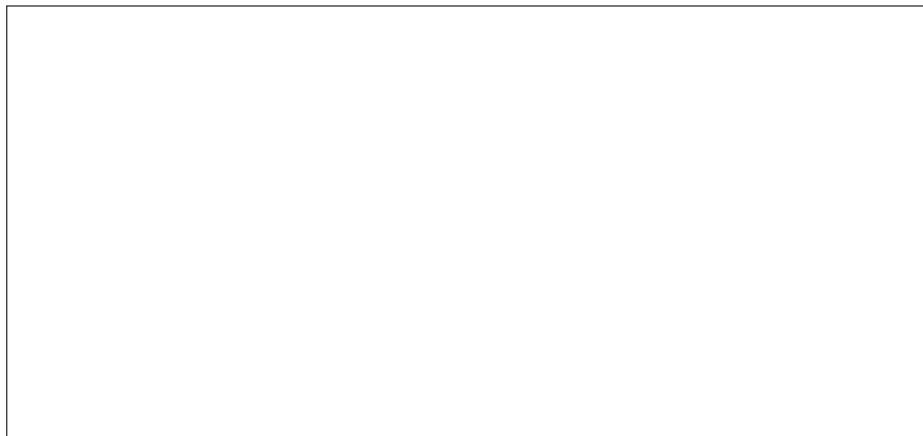


Figure 6: Voltage at node P (lower trace) and output U_{OUT} (upper trace) – two-inverter oscillator.

Hints for this section

- First inverter output feeds back through R - C to its own input at node S
- When U_{OUT} switches high, C begins charging through R ; voltage at S rises exponentially until it crosses the inverter switching threshold
- Second inverter buffers and re-inverts the signal
- R_S limits current when V_S overshoots V_{CC} or undershoots GND
- Output is an approximate square wave; node P shows an exponential waveform clipped by the CMOS input protection diodes

[Describe what is observed on the oscilloscope at U_{OUT} and P. State the observed frequency.]

5.3.2 Task 1.2 – Measurement at S and P

Table 7: Measured values – two-inverter astable oscillator.

Quantity	Symbol	Calculated	Measured
Oscillation frequency	f	45.5 kHz	?? kHz
Period	T	22.0 μ s	?? μ s
Voltage swing at S	ΔV_S	$\approx V_{CC}$	?? V
Voltage swing at P	ΔV_P	$> V_{CC}$	?? V
Peak voltage at P	$V_{P,\max}$	—	?? V
Valley voltage at P	$V_{P,\min}$	—	?? V

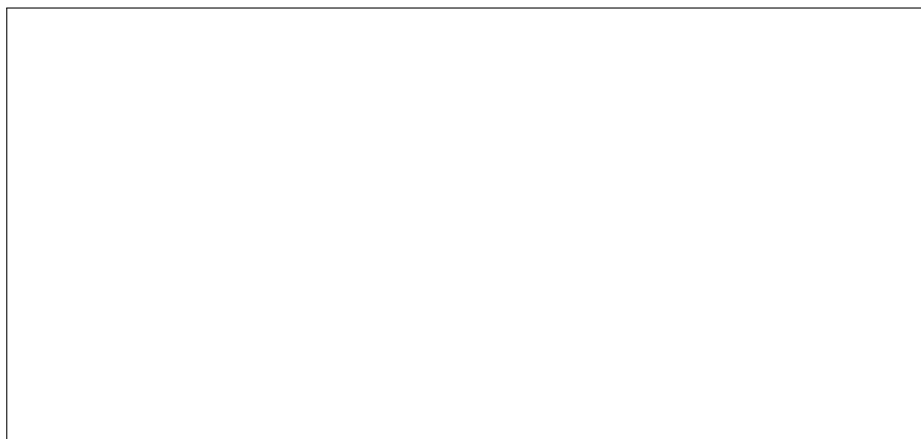


Figure 7: Waveforms at S (upper trace) and P (lower trace) – two-inverter oscillator.

Hints for this section

- S: square wave switching between 0 V and V_{CC}
- P: exponential charging/discharging, clamped by CMOS protection diodes to approximately -0.7 V and $V_{CC} + 0.7$ V
- The CMOS switching threshold is typically at $\frac{1}{2}V_{CC}$
- The clipping of the P waveform is clearly visible on the oscilloscope

[Describe both waveforms in detail. Explain why the voltage at P exceeds the supply rails and how the protection diodes limit it.]

5.3.3 Task 1.3 – Purpose of R_S

Hints for this section

- Without R_S : unlimited current through the CMOS input protection diode when V_P exceeds V_{CC} or drops below GND – IC damage possible
- R_S limits diode current to a safe value: $I_D \approx \frac{V_{P,\max} - V_{CC}}{R_S}$
- R_S also slightly reduces the oscillation frequency (adds to R)
- Typical recommendation: $R_S \geq 1$ k Ω for 74C-series CMOS

With $R_S = 1\text{ k}\Omega$, the maximum protection diode current is limited to:

$$I_{D,\max} \approx \frac{V_{P,\max} - V_{CC}}{R_S} = \frac{??\text{ V} - 10\text{ V}}{1\text{ k}\Omega} = ??\text{ mA}$$

[Explain in your own words why R_S is necessary and what would happen without it.]

5.4 Task 3.4 – Quartz Crystal Oscillator (74C00, 12 MHz)

Component values used: $R_1 = 1\text{ M}\Omega$, $R_2 = 220\text{ }\Omega$, $C_1 = C_2 = 68\text{ pF}$, quartz: 12 MHz, $V_{CC} = 5\text{ V}$.

5.4.1 Task 1.1 – Output Voltage Measurement at $V_{CC} = 5\text{ V}$

Table 8: Measured values – quartz crystal oscillator at 5 V.

Quantity	Symbol	Expected	Measured
Frequency	f	12.000 MHz	?? MHz
Period	T	83.33 ns	?? ns
Peak-to-peak voltage	V_{pp}	$\leq V_{CC}$	?? V
Rise time	t_r	—	?? ns

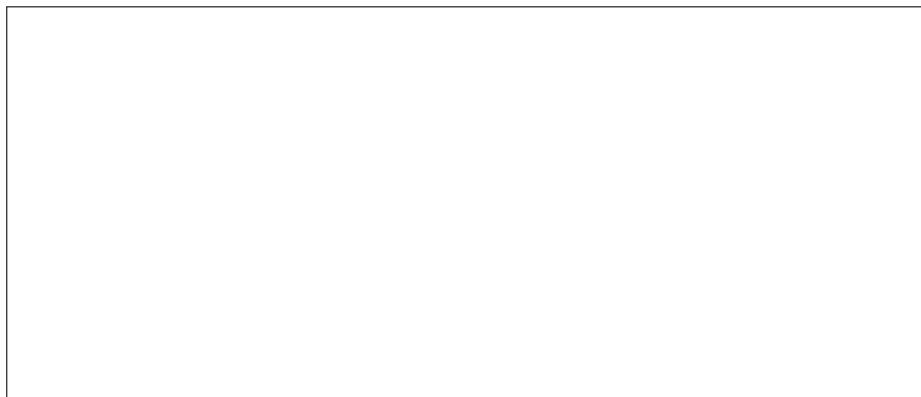


Figure 8: Output waveform of the quartz crystal oscillator at $V_{CC} = 5\text{ V}$.

5.4.2 Task 1.2 – Explanation of the Curve

Hints for this section

- Quartz crystal: very high Q-factor ($Q \sim 10^4\text{--}10^6$), acts as a narrow bandpass filter in the feedback path
- R_1 biases the inverter into its linear (amplifying) region
- C_1 , C_2 and the quartz form the Pierce oscillator topology
- Frequency determined almost entirely by the quartz – not by R or C values
- R_2 damps parasitic oscillations and protects the output stage from excessive current
- Waveform near the crystal node may appear nearly sinusoidal due to filtering

[Describe the waveform shape (sinusoidal, square, or intermediate?). Comment on the frequency accuracy compared to the quartz nominal frequency.]

5.4.3 Task 1.3 – Effect of Increased Supply Voltage

Table 9: Output behaviour at different supply voltages – quartz oscillator.

V_{CC}	f (MHz)	V_{pp} (V)	Waveform shape	Observations
5 V	??	??	<i>[shape]</i>	Nominal operation
10 V	??	??	<i>[shape]</i>	<i>[describe]</i>
15 V	??	??	<i>[shape]</i>	<i>[describe]</i>

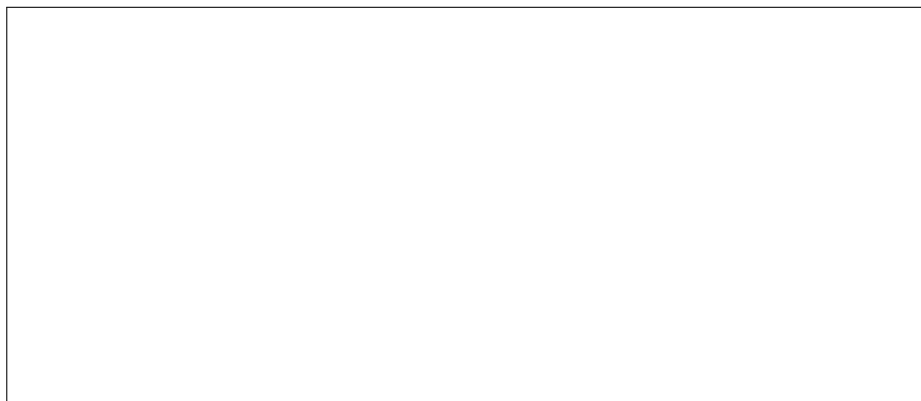


Figure 9: Quartz oscillator output at 5 V, 10 V, and 15 V.

Hints for this section

- 74C00 absolute maximum V_{CC} : check datasheet (typically 18 V)
- Higher V_{CC} : faster propagation delay, hence slight frequency increase
- Higher V_{CC} : larger V_{pp} at output
- Risk of IC damage above absolute maximum rating
- Quartz resonant frequency is nearly independent of V_{CC} , but very sensitive to temperature

[Describe and explain the effect on frequency, amplitude, and waveform shape. Did the circuit still oscillate at all supply voltages tested?]

6 Conclusions

The following table provides a concise overview of calculated vs. measured values for all tasks:

Table 10: Summary: calculated vs. measured values for all tasks.

Task	Quantity	Calculated	Measured	Rel. error
3.1	t_H (monostable)	1.100 s	?? s	?? %
3.2	T (astable NE555)	2.079 ms	?? ms	?? %
3.2	f (astable NE555)	481 Hz	?? Hz	?? %
3.3	f (inverter osc.)	45.5 kHz	?? kHz	?? %
3.4	f (quartz osc.)	12.000 MHz	?? MHz	?? %

Hints for this section

- Compare all calculated vs. measured values from the table above
- Discuss the dominant error source for each circuit
- Comment on the probe vs. BNC measurement (task 3.1.4)
- State whether all experiment objectives were achieved

[Write 4–6 sentences summarising the main findings. Refer to Table 10.]

7 Summary

[3–5 sentence recap: restate the objective, briefly name the methods, state the key numerical results, and give the main conclusion. No new information may appear here. Do not copy from the Introduction – paraphrase.]

8 Outlook

[Suggest 2–3 possible follow-up experiments or improvements, for example: duty-cycle adjustment of the astable NE555 using a diode in parallel with R_2 ; temperature dependence of the RC-based oscillators; comparison of quartz vs. ceramic resonator frequency stability.]

References

- [1] T. Floyd, *Digital Fundamentals*, 9th ed. Pearson Prentice Hall, 2006, ch. 7-6.
- [2] U. Tietze and C. Schenk, *Halbleiterschaltungstechnik*, 11. Auflage. Springer, 1999, Kap. 6.5.
- [3] C. Thorrold, *Guide for Writing Lab Reports*, Version 1.0.
- [4] Texas Instruments, *NE555 Precision Timers – Datasheet*, <https://www.ti.com/lit/ds/symlink/ne555.pdf>, accessed April 2026.
- [5] Texas Instruments, *CD74C00 Quad 2-Input NAND Gate – Datasheet*, <https://www.ti.com/lit/ds/symlink/cd74hc00.pdf>, accessed April 2026.

A Appendix A – Raw Oscilloscope Screenshots

[Insert full-resolution screenshots here, labelled with task number, channel settings (V/div, time/div), and trigger settings.]

B Appendix B – Actual Component Values

Table 11: Component values measured with multimeter before assembly.

Component	Nominal	Measured	Tolerance	Task
R_1 (monostable)	1 M Ω	?? MΩ	$\pm 5\%$	3.1
C (monostable)	1 μ F	?? μF	$\pm 20\%$	3.1
R_1 (astable)	10 k Ω	?? kΩ	$\pm 5\%$	3.2
R_2 (astable)	10 k Ω	?? kΩ	$\pm 5\%$	3.2
C (astable)	100 nF	?? nF	$\pm 20\%$	3.2
R_S (inverter)	1 k Ω	?? kΩ	$\pm 5\%$	3.3
R (inverter)	10 k Ω	?? kΩ	$\pm 5\%$	3.3
C (inverter)	1 nF	?? nF	$\pm 20\%$	3.3